INCH-POUND
MIL-DTL-20625B
27 August 2013
SUPERSEDING
MIL-U-20625A
28 March 1957

DETAIL SPECIFICATION

UNIVERSAL JOINTS, BLOCK TYPE, MECHANICAL REMOTE CONTROL SYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers universal joints of the block type for use in rotary type hand-operated or intermittent power-operated valve remote control systems of the mechanical type.

1.2 Classification. Universal joints are of the following classes, as specified (see 6.1 and 6.2):

Class A – Steel

Class B – Bronze

Class C – Steel, Corrosion-Resistant

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:07Z
Check the source to verify that this is the current version before use.

ASTM INTERNATIONAL

ASTM A304	-	Standard Specification for Carbon and Alloy Steel Bars Subject to End-Quench Hardenability Requirements
ASTM A582/A582M	-	Standard Specification for Free-Machining Stainless Steel Bars
ASTM B21/B21M	-	Standard Specification for Naval Brass Rod, Bar, and Shapes

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The materials for parts shall be in accordance with table I, as applicable. Materials for parts not covered by table I shall be of high grade commercial standard material which, when installed in a mechanical valve remote control system and operated intermittently, will transmit the required torques specified in table II (see 4.1.2).

TABLE I. Materials.

	Hubs	Block	Pins
Class A	ASTM A304	ASTM A304	ASTM A304
Class B	ASTM B21/B21M	ASTM B21/B21M	QQ-N-281
Class C	ASTM A582/A582M	ASTM A582/A582M	ASTM A582/A582M

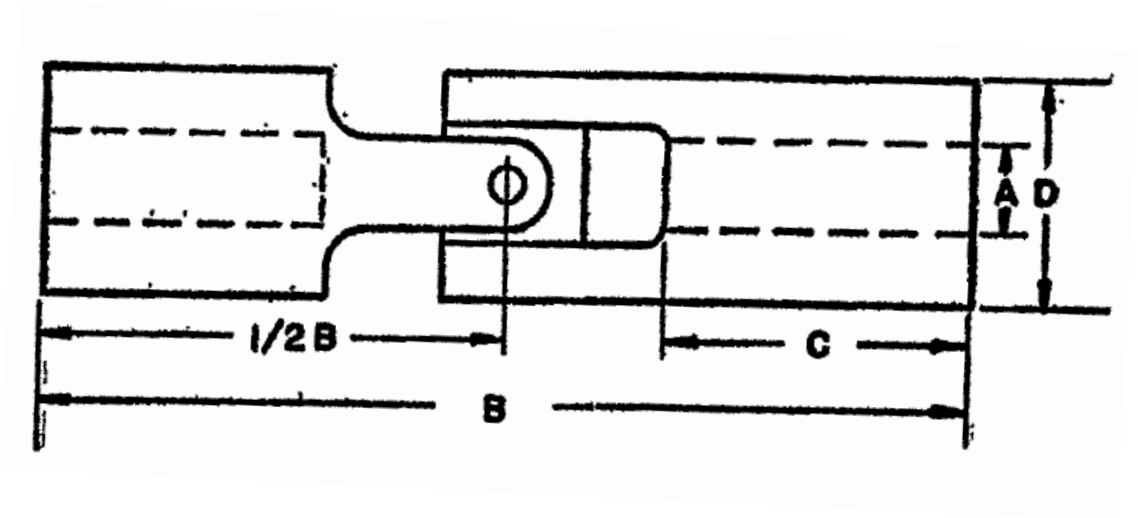
TABLE II. Transmitted torque (foot-pounds).

Joint Size	150 Degrees	180 Degrees
⅜	2	2.5
½	5.6	7
⅝	12	15
¾	14	-17.5
⅞	16	20
1	32	40
1⅛	41.6	52
1¼	56	70
1½	-68	105
1¾	80	125
2	130	180
2½	200	250
3	300	375
4	400	500

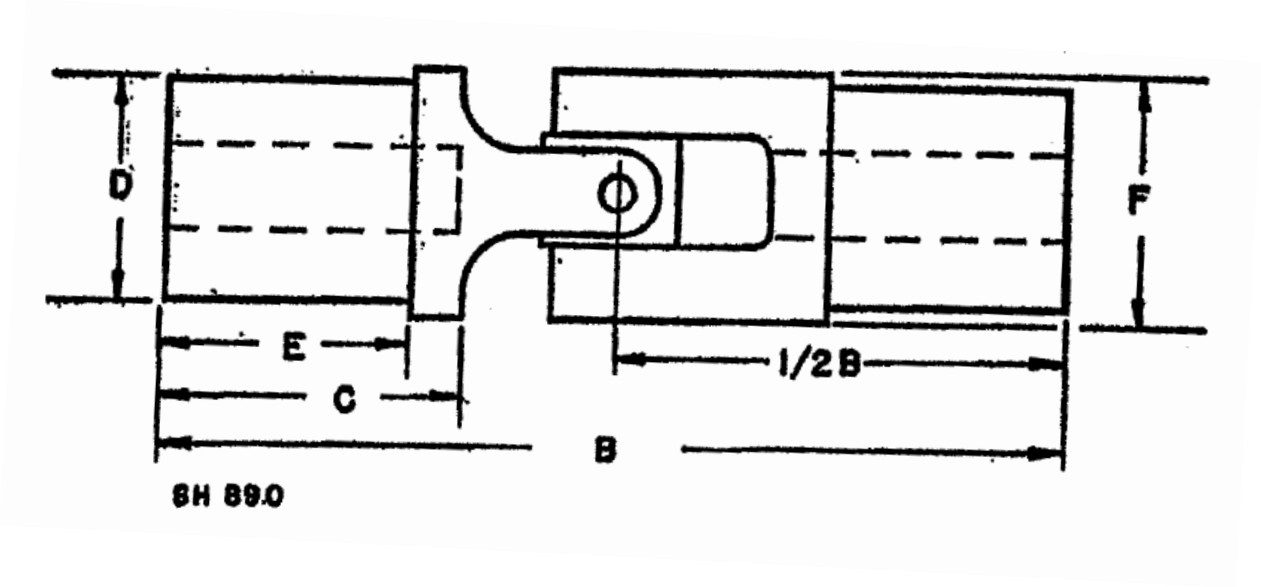
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:07Z
Check the source to verify that this is the current version before use.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 General design. Universal joints shall consist of two hubs with extended jaws, herein called the "yoke", block and bearing pins, assembled so that a torque applied to one hub will be transmitted to the other. The dimensions shall conform to table III and joints shall be assembled as shown on figure 1 (see 4.2.1).



Classes A and C



Class B

FIGURE 1. Universal joints.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:07Z
Check the source to verify that this is the current version before use.

TABLE III. General dimensions (inches).

A, Shaft Size (max.)	B, Overall Length		C, Hole Length		D, Hub Diameter		E, Hub Length Class B	F, Offset Diameter Class B
	Classes A and C	Class B	Classes A and C	Class B	Classes A and C	Class B		
3/16	1 3/4	4 3/4	11/16	1 1/2	3/8	1 1/4	1 1/8	1 3/8
1/4	2	4 3/4	3/4	1 1/2	1/2	1 1/4	1 1/8	1 3/8
5/16	2 1/4	4 3/4	13/16	1 1/2	5/8	1 1/4	1 1/8	1 3/8
3/8	2 11/16	4 3/4	31/32	1 1/2	3/4	1 1/4	1 1/8	1 3/8
1/2	3	4 3/4	1 1/32	1 1/2	7/8	1 1/4	1 1/8	1 3/8
5/8	3 3/8	4 3/4	1 1/8	1 1/2	1	1 1/4	1 1/8	1 3/8
9/16	3 1/2	4 3/4	1 7/32	1 1/2	1 1/8	1 1/4	1 1/8	1 3/8
3/4	3 3/4	4 3/4	1 3/16	1 1/2	1 1/4	1 1/4	1 1/8	1 3/8
7/8	4 1/4	4 3/4	1 11/32	1 1/2	1 1/2	1 1/4	1 1/8	1 3/8
1	5	6 3/4	1 9/16	2 1/4	1 3/4	1 3/4	1 3/4	2
1 1/4	5 7/16	8	1 5/8	2 1/2	2	2 1/4	1 7/8	2 3/4
1 1/2	7	8	2 5/32	2 1/2	2 1/2	2 1/4	1 7/8	2 3/4
1 3/4	9	9	2 11/16	2 3/4	3	2 3/4	2 1/8	3 1/8
2	10 5/8	----	3 1/8	----	4	----	----	----

3.3.1 Sizes. Universal joints shall be designated by the outside diameter of the hub (see D of table III). For a given shaft size, the joint shall be equal to or larger than that in table III, as specified (see 4.2.1 and 6.2).

3.3.2 Hubs. Unless otherwise specified (see 6.2), the joints shall be furnished with the hubs machined with centers but not drilled for shafting. When specified (see 6.2), the diameter and length of the hubs shall conform to table III for the shaft size indicated (see 4.2.1).

3.3.3 Bearing pins. The bearing pins, when assembled with the hubs and block, shall not project outward beyond the yoke and shall be positively secured in operating position so that operation will not cause the pins to become loose or drop out (see 4.2.1).

3.3.4 Dimensional tolerances. Clearance permitted between pins and holes shall be 0.001 inch maximum. Clearance between block and yoke shall be 0.004 inch maximum. Pins may be off-center 0.005 inch maximum. All other dimensions shall be ±0.020 inch (see 4.2.1).

3.3.5 Lubrication. An oil passage shall be provided in one of the pins through which a lubricant may be applied to the working surfaces (see 4.2.1).

3.3.6 Fastenings. Set screws shall not be used in universal joints (see 4.2.1).

3.4 Performance. The universal joint shall be capable of transmitting the static torque shown in table II at the indicated angle without rupture, permanent set, or other evidence of failure (see 4.3.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:07Z
Check the source to verify that this is the current version before use.

3.4.1 Backlash. The universal joint shall exhibit no more than the maximum backlash set forth in table IV (see 4.3.1).

TABLE IV. Maximum backlash.

Hub Outside Diameter	Backlash (Degree)
3/8 – 1	0.6
1 1/8	0.5
1 1/4	
1 1/2	
1 3/4	
2	
2 1/2	0.3
3	
4	

3.4.2 Efficiency. The universal joint shall transmit the torques shown in table II with an efficiency of not less than 98 percent with the axes of the hubs at 180 degrees, and 80 percent with the joint at an angle of 150 degrees between the axes of the hubs (see 4.3.2).

3.5 Identification marking. Each joint shall be permanently and legibly marked with the manufacturer's name, trademark, or symbol in accordance with MIL-STD-130 (see 4.2.1).

3.6 Surface roughness. All nonworking-machined surfaces shall have 125-microinch finishes or better, bearing surfaces shall have 32-microinch finishes or better (see 4.2.1).

3.7 Workmanship. Workmanship shall be in accordance with high grade commercial practice governing this type of material. Each universal joint shall be uniform in quality and temper and shall be free from burrs, silvers, gouges, porosity, cracks, objectionable scale, or any other defects which may adversely affect serviceability (see 4.2.1).

4. VERIFICATION

4.1 Sampling.

4.1.1 Lot. All joints of the same class and size presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.1.2 Sampling for visual and dimensional examination. A random sample of joints shall be selected from each inspection lot of material offered for Government inspection in accordance with MIL-STD-1916 at Inspection Level III for lots of 170 and under and Inspection Level II for lots over 170.

4.2 Lot acceptance inspection and tests.

4.2.1 Visual and dimensional examination. Each of the sample units selected in accordance with 4.1.2 shall be visually and dimensionally examined to verify compliance with this specification. Any unit in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number as specified (see 6.2) for that sample, the lot represented by the sample shall be rejected.

4.2.2 Tests. Each of the sample joints selected as specified in 4.1.2 shall be subjected to the tests specified in 4.3 to determine compliance with this specification. Any unit in the sample containing one or more defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number as specified (see 6.2) for the sample, the lot represented by the sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:07Z
Check the source to verify that this is the current version before use.

4.3 Test procedures.

4.3.1 Backlash. The backlash of the joints shall not exceed the amount specified in table IV when the joint is mounted in a test apparatus with the hubs at 180 degrees. The following details shall apply:

a. Apply a torque of 3.5 to 4.5 inch-pounds for joint sizes 3/16 through 3/4inch or 7.5 to 8.5 inch-pounds for all other sizes to the universal joint to obtain the initial reading.

b. The torque load shall then be reversed and a second reading taken.

c. The backlash at the circumference of the universal joint is the difference between the two readings.

4.3.2 Torque. The joint shall be subjected to the torques specified in table II at four or more radial positions of the hub and at the angles specified. The load shall be sustained at each position for 3 minutes.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Class A universal joints are intended for general purpose use in mechanical remote control systems. Class B joints are intended for use where corrosive conditions are abnormal or nonmagnetic materials are required. Class C joints are intended for use where corrosive conditions are abnormal or high torque requirements are desired.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class required (see 1.2).

c. Number of each size required; designated by hub outside diameter (see 3.3.1).

d. Shaft size (see 3.3.2).

e. Acceptance number of defective units (see 4.2.1 and 4.2.2).

f. Packaging requirements (see 5.1).

6.3 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmentally Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website: http://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the EPA list of 31 priority chemicals are cadmium, lead, and mercury. Use of these materials on the list should be minimized or eliminated unless needed to meet the requirements specified herein (see 3.2).

6.4 Subject term (key word listing).

Bearing pin

Hub

Torque

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:07Z
Check the source to verify that this is the current version before use.

Custodians:
Army – AT
Navy – SH
DLA – CC

Preparing Activity:
Navy – SH
(Project 3010-2013-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:07Z
Check the source to verify that this is the current version before use.